Project 2 - Web Testing Selenium

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Selenium

- A suite of tools for automating web browsers
- First launched at 2002/04/30
- Written in Java
- A cross platform software
- Support programming languages
    - C#, Javascript, Java, Python, Ruby

Selenium IDE

- A browser extension that records and plays back a user's actions
- Web ready
- Easy debugging
- Cross-browser execution

Selenium WebDriver

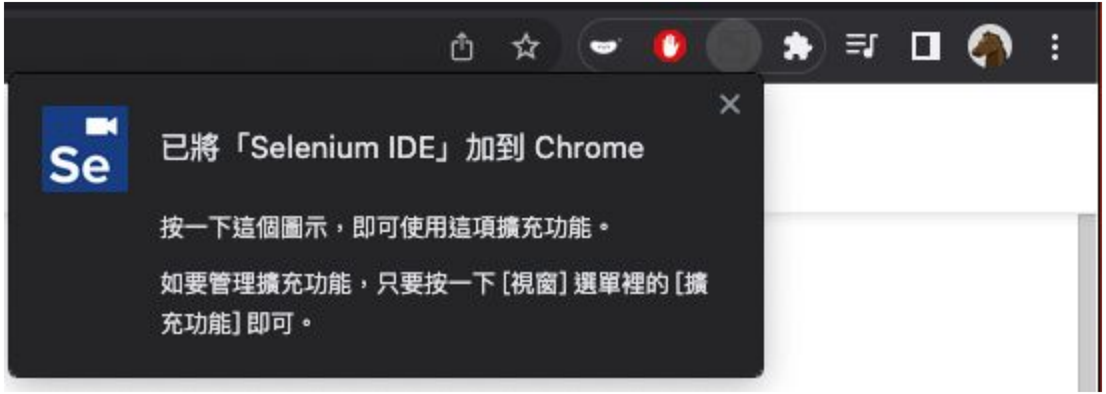
- Designed as a simple and more concise programming interface

Selenium Grid

- Provide an easy way to run tests in parallel on multiple machines
- Allow testing on different browser versions
- Enable cross platform testing

Installation (Selenium IDE)

- Install Selenium IDE extension on Chrome/FireFox
  - use Chrome here
  - https://chrome.google.com/webstore/detail/selenium-ide/mooikfkahbdckldjjndioackbalphokd

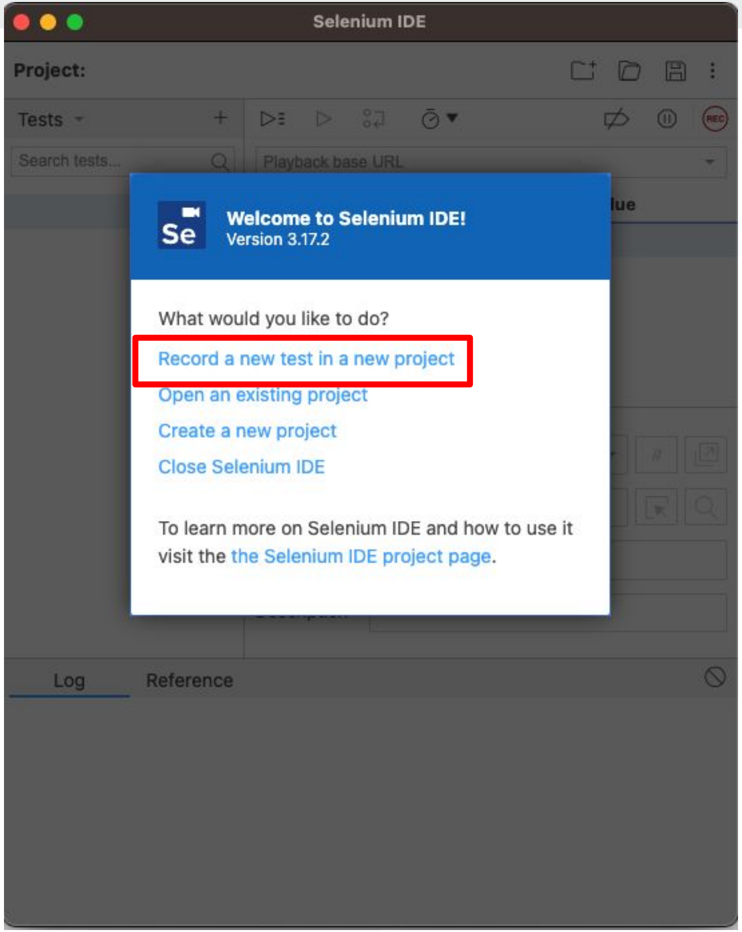


Start with Selenium IDE

1. Create a new project / Open an old project
2. Add a new test to project
3. For a single test in a project
    a. Recording
    b. Insert and Delete commands
    c. Playback
    d. Export to programming language
4. Save the project into .side file

Record a Test

- Record your actions and save as a list of commands
- e.g.
  - open https://www.google.com/
  - type 'facebook' in search bar
  - press enter
  - click 'Facebook' page
  - close the page



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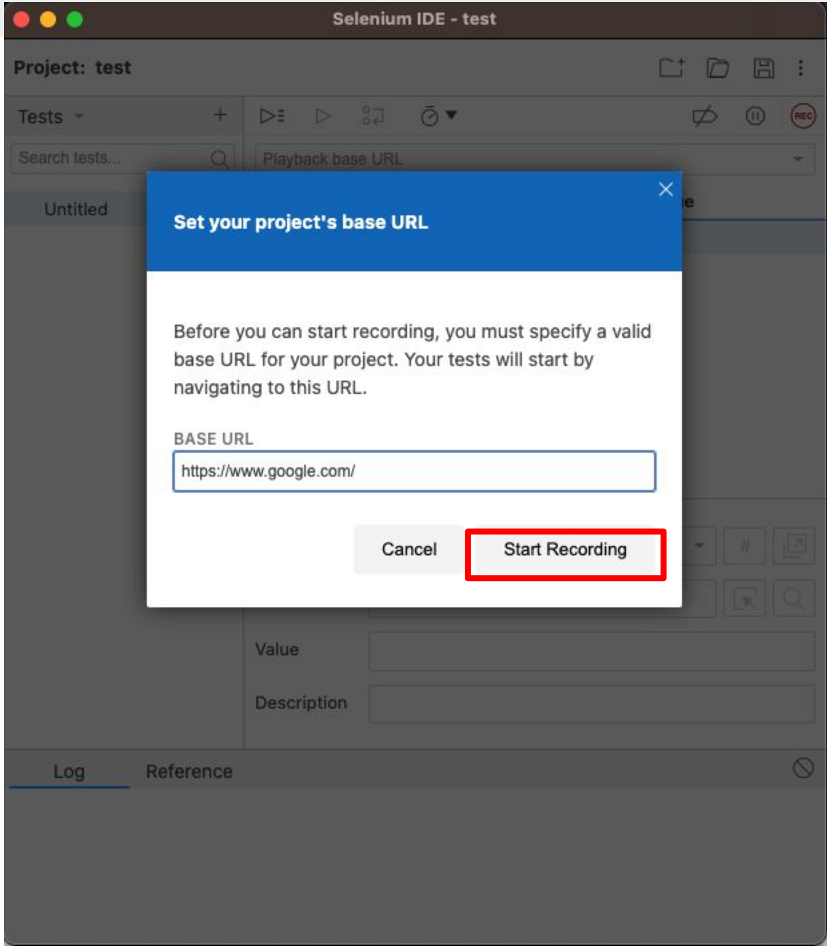

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❏ Right click on the page

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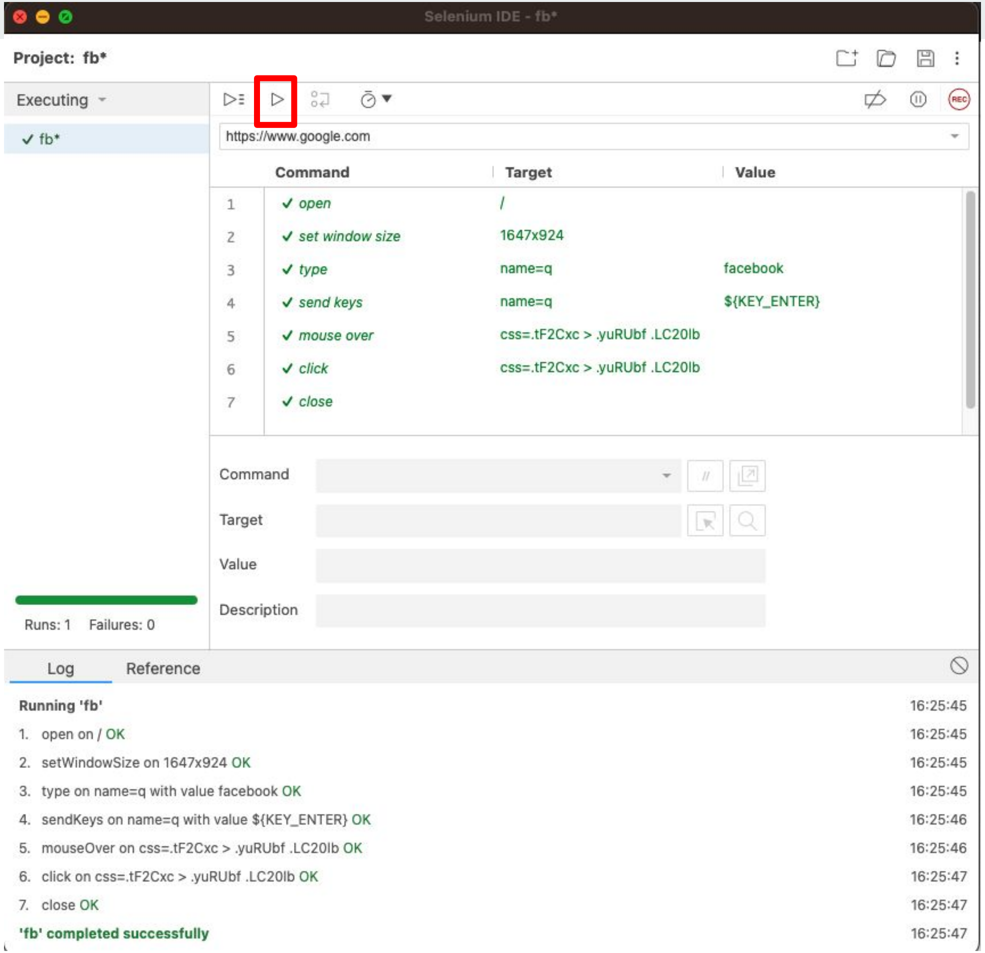
	Command	Target	Value
1	open	/	
2	set window size	1647x924	
3	type	name=q	facebook
4	send keys	name=q	${KEY_ENTER}
5	mouse over	css=.tF2Cxc > .yuRUbf .LC20lb	
6	click	css=.tF2Cxc > .yuRUbf .LC20lb	
7	close		

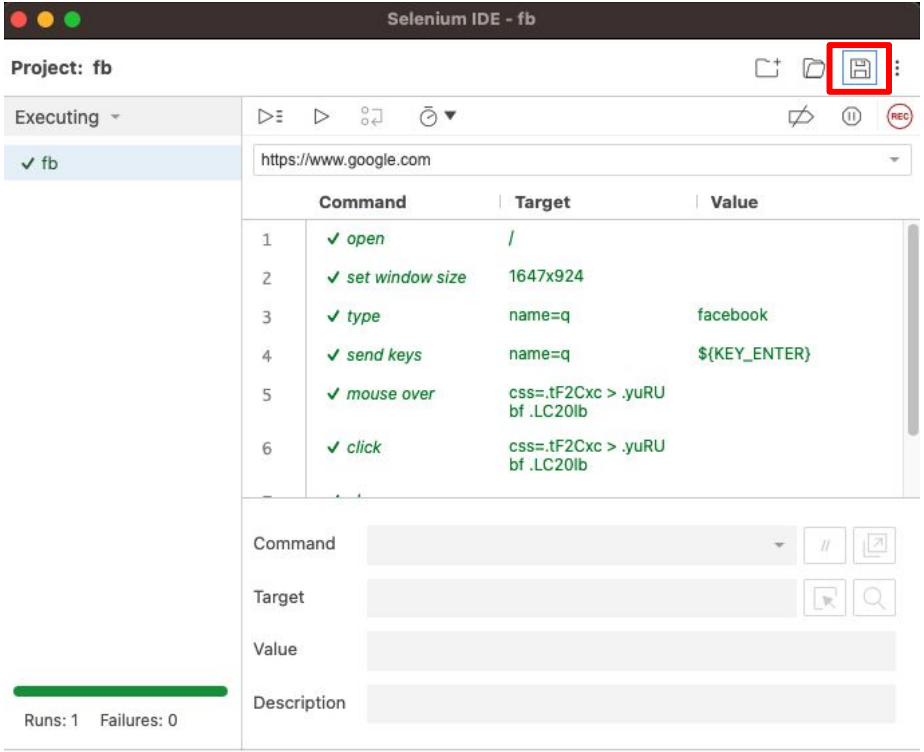
Run a Test

- Run the recorded actions
- Selenium IDE provides result of running with logs



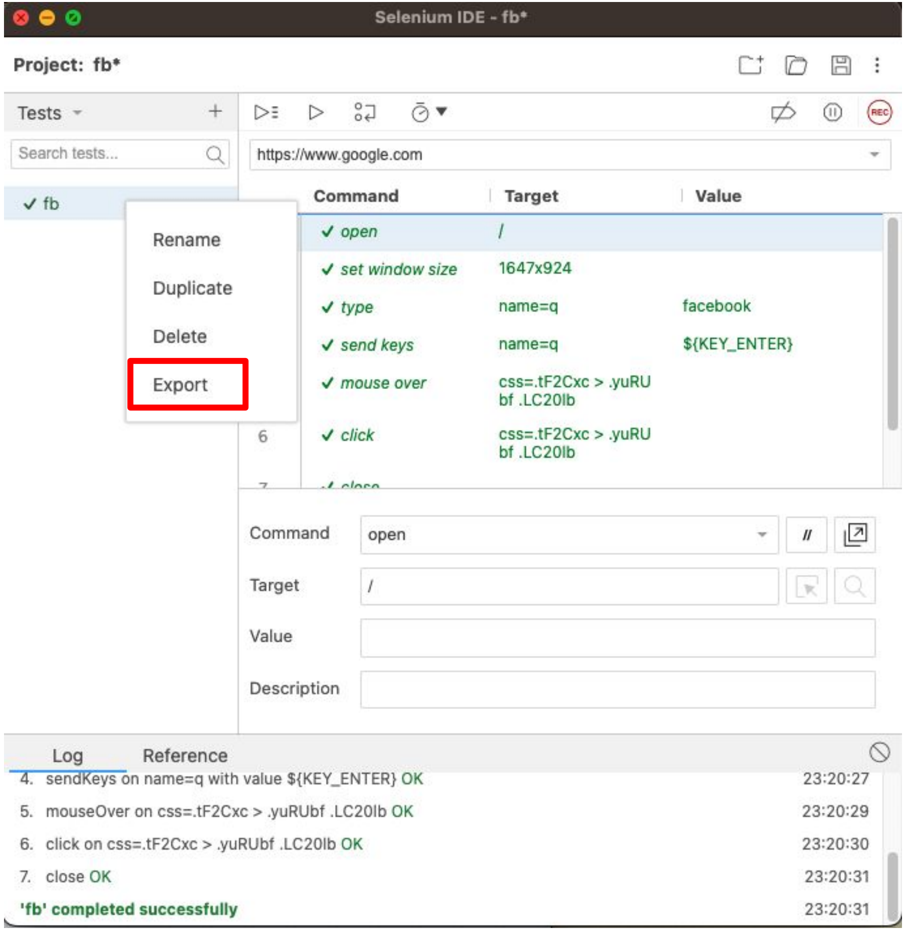
Save a Test to Project

- Save the recorded actions to .side file



Export a Test

- Export the recorded actions into programming language



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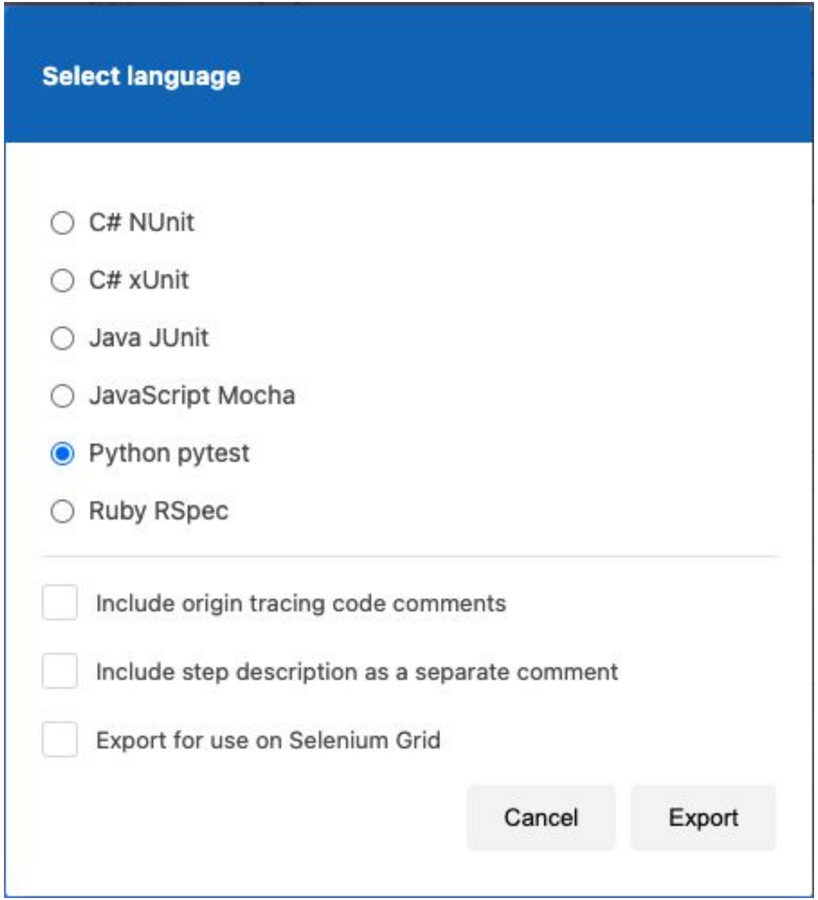

Requirement & Evaluation of Project2

- Find one website with multiple language choices
  - Register the website different from other students
  - https://docs.google.com/spreadsheets/d/13_GNUfqFi0Wk2N2mh-_DcKhDZwMYpm6iAKFxQoCV_gk/edit?usp=sharing
- (10%) Use assertions to test whether the contents change appropriately according to language selection
  - (9%) The navigation bar
  - (1%) Webpage title
- (5%) Pick two other pages in the same website and test if language selection is properly remembered
- Total 10% + 5% = 15%

Requirement & Evaluation of Project2 (cont.)

- Write a report specifies
  - the webpage you choose
  - test script you design
  - testing result

Submission

- Hand in the following files to NTU COOL
  - your testing script (.side)
  - report
- Deadline: 2023/5/3 23:59